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# A Sustainable Solid Waste Management System: An Integrated Approach to Transform Lucknow into a Sustainable City

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## Abstract

In urban areas, solid waste management is a very difficult task for conserving our natural environment conservation across the globe. Lucknow Municipal Corporation (LMC) is the 10th largest solid waste generating metropolitan in the country produces a total of 1500 tons of solid waste per day, and the average per-capita per day solid waste generation is 0.65 kilogram during 2021. LMC is unable to manage such a huge amount of solid waste which is growing the number of dumping grounds and garbage mounting at different locations in the city. The present research article is an attempt to study solid waste management and presents a prospect of a sustainable approach for managing municipal solid waste management in the coming future to transform Lucknow into a sustainable city. The integrated solid waste management (ISWM) approach is the best approach for managing solid waste along with a strategy of extended producer responsibility model of a circular economy. SSWMS is a comprehensive approach that applies the four R's of recycle, reduce, reuse, and refuse to have a sustainable waste management system. The research article attempted to explain the problem of solid waste management in the city in the future. Eventually, the concluding remarks advocate an innovative method to manage solid waste materials contributing to achieving the city's sustainable development goal.

## INTRODUCTION

Solid waste management is a very task to conserve the natural environment across the globe, especially in urban areas. The United Nations has fixed the Agenda 2030 for sustainable development, accepted and adopted by all its members in 2015. The agenda 2030 includes the 17 sustainable development goals (SDGs) in a global partnership. The 11$^{th}$ goal of the 2030 agenda of SDGs focuses on sustainable cities and communities to make the cities and human settlements inclusive, safe, resilient, and sustainable. India is one of the UNs' countries that adopted 17 SDGs to fulfill the goals of sustainable development in India. On this line, the Indian government has proposed major Indian cities transform into sustainable cities. Lucknow is India's eleventh-most populous city and the twelfth-most populous urban agglomeration. It is the largest capital city and the second-largest urban agglomeration in the state of Uttar Pradesh. As per the Census of India, Lucknow is the administrative district headquarters and the division has a 2.8 million population. The current estimated population of Lucknow city in 2023 is 3,884,000. Lucknow municipal corporation (LMC) is India's 10th largest municipal solid waste-generating metropolitan city, producing a huge amount of solid waste daily. Resultantly such a huge amount of solid waste is constantly accumulating the number of dumping grounds and garbage mounting at different locations in Lucknow city. Therefore, the present article attempts to study the solid waste management system working in the city. This article is based on secondary data mainly availed from the Lucknow Municipal Corporation (LMC). The research article has attempted to measure the magnitude of the city's solid waste management problem. Finally, the article advocates an innovative method to manage solid waste materials contributing to achieving the goal of sustainable development.

### Statement of the Problem

Lucknow District's population constitutes 2.30 percent of the total Uttar Pradesh population (Census of India, 2011). Around 1,550,737 people are living in rural areas and 3,037,718 in urban areas. As per United Nations, World Population Prospects, the current (2023) population of the Lucknow metropolitan area is 3,945,000, which has increased a 2.36% from 2022.

An estimate by United Nations, World Population Prospects, shows that the total population of the Lucknow metropolitan area will be increased (2.0–3.4%) by 2031 and will reach 4731000 in 2031 (Figure 1). As per the estimate of lucknow development authority (LDA) Master Plan 2031, less than 40% city is urbanized. The urbanized area of Lucknow city will increase by up to 70% by 2031. The total area of Lucknow municipality area was 450 km$^2$ has now enlarged to more than

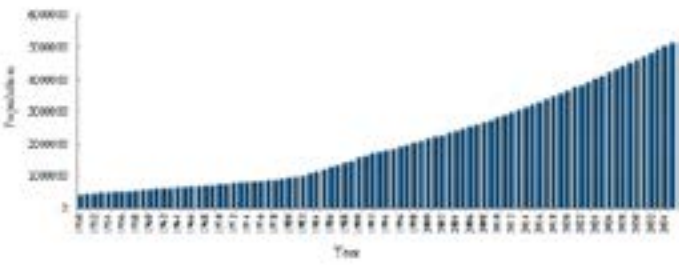


**Figure 1:** Annual Increase of Population, Lucknow city (1950-2035)
**Data Source:** World Population Prospects, United Nations

over 1,050 square kilometers. As per Lucknow Development Authority (LDA) estimate, Lucknow municipality's population will rise to 65 lacks by 2031 which will enlarge the boundaries of the city.

Lucknow city is spread over an area of about 247.7 square kilometers where, total solid waste generation is approximately 1500 Metric tons/day in 2011 (Lucknow Nagar Nigam, 2011). Lucknow Municipal Corporation (LMC) is the 10th largest municipal solid waste (MSW) generating metropolitan city in India that produce a total of 1500 ton of solid waste per day, and the average per-capita per day solid waste generation is 0.65 kilogram per person in 2021 (LMC, 2021). LMC is unable to manage such a huge quantity of solid waste which is increasing the number of dumping grounds and garbage mounting at different locations in Lucknow city (LMC, 2021).

The Lucknow city development plan 2006 shows that there are three major problems related to solid waste management i.e., i) inadequate infrastructure, ii) inefficient Solid Waste Management system, and, iii) inefficient and inadequate institution. There are insufficient numbers of infrastructure facilities i.e., solid waste depots in Lucknow city. Moreover, these depots are located near the primary collection points which allow for the disposal of wastes in the open, increasing pollution. Additionally, maintenance of vehicles and equipment is inadequate and inefficient which fails the solid waste management system of the city. Lucknow waste management system is inefficient i.e., rag pickers, swappers, toilet cleaners, etc. are to health risks and spread health risks further. The Lucknow waste management system's activities are unorganized and unsystematic. The waste depots are not unsanitary and, during rains, become large pools of leachates, breeding grounds for different types of diseases. There are insufficient numbers of staff, vehicles and equipment, rag pickers, swappers, toilet cleaners, etc. Additionally, inefficient, weak coordination within the primary collection and secondary transport, and weak and ineffective cooperation between the labor force and management makes the task more challenging (Lucknow City Development Plan, 2006). It is therefore revealed that if the urban population of Lucknow metropolitan city will increase as it has predicted (Figure 2) above; the area of the city will also enlarge in the same proportion will accelerate the rate of solid waste generation. Resultantly, problems of waste management will be very serious from a different point of view i.e., pollution, speared of different types of diseases, hygiene, sanitation, etc. There is, therefore, a need to design an innovative and efficient waste management system to conserve our natural environment safely and healthily and transform Lucknow into a smart and sustainable city.

It has been noted that the Lucknow municipal corporation is unable to manage the generated solid waste properly. The study's key objective is to prospect a sustainable approach to managing solid waste for transforming Lucknow into a sustainable city.

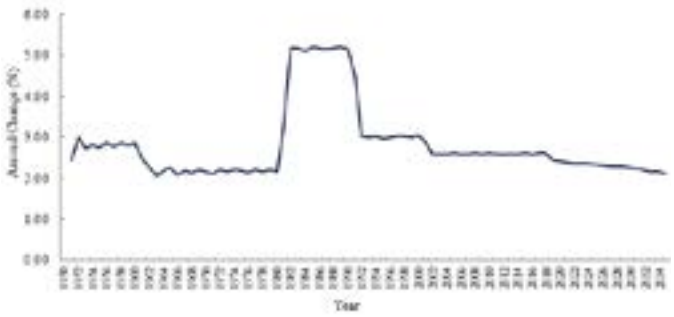



**Figure 2:**Annual change in population, Lucknow city (1950-2035)
**Data Source:** World Population Prospects, United Nations

## MATERIALS AND METHODS

The present research work is a quantitative exercise based on secondary data extracted from Lucknow Municipal Corporation (LMC) portal, Master Plan 2021, Census of India, World Population Prospects of United Nations and government reports, and other sources. The article is an attempt to address the problem of solid waste management system in Lucknow city at present and in the coming future till 2030. Further, it provides intellectual discourse based on different types of available literature on the subject matter. A map of the study area has been created using ArcGIS 10 Software of ESRI. Besides, bar-digraph and figures have been prepared on the basis of the results of the available data for clarifying the issue to fulfill the study's objective.

### Lucknow Municipal Corporation (LMC)

Lucknow city is the capital of Uttar Pradesh, located on the bank of the Gomti River at 26° 51' 0" North Latitude and 80° 55' 0" East Longitude, covering an area of about 450 square kilometers. Lucknow city is located at a height of 123 meters above the mean sea level. As per the Census of India 2011, the total population of Lucknow is 2,817,105. However, Lucknow's metropolitan population is 2,902,920 of which the male population constitutes 52% and female shares 48% of the total. The Lucknow Municipal Corporation has divided the whole city into 110 wards and 7 zones for managing solid waste (City Development Plan Lucknow, 2021).

LMC is responsible for collecting, transporting, treating, and disposing of the city's generated solid waste. There was 1500 metric tons of solid waste generation per day in Lucknow city (LNN, 2011). As per the municipal commissioner office of Lucknow, 'Lucknow produces about 2,000 metric tons of solid waste per day; out of that half of the semi-solid waste can be turned into manure (Chauhan, 2022).

Lucknow city's average per capita per day production of Solid Waste (OECD, 2013) is 0.65 kilograms. As per Nagar Nigam Lucknow, there are 52 ragbags dumping depots in the Lucknow. The LMC works for collecting solid waste and deposition it at the depots. After that, solid waste is transported to various landfills and other processes to manage in the city. Indian cities are experiencing rapid economic growth for the last few decades, leading to accelerated consumption and waste generation rate. Furthermore, economic growth usually leads to the consumption of innovative goods like sophisticated electronic devices which are very hard to recycle. Moreover, population growth speeds up the waste generation rate.




**Lucknow Map 1**

Most often such expansion of population occurs in thickly populated zones of the cities that worsen difficulties to collect waste. In other words, solid waste generation is directly related to the city's population and the city's size. As the size of the city expands and so enlarges, the population of the urban centers. Social and economic activities generate different types of waste from different sectors in different types as byproducts of human activities are proportional to the size of the population. Therefore, we have tried to analyze the increasing urbanization and enlarging population in Lucknow city.

**Population**

The city's population pattern is an important factor in determining the amount of solid waste generation and its management. Population growth is directly correlated with the amount of solid waste generation. However, rapid population growth primarily caused urbanization, and industrialization contributes to steadily generating solid waste (Hiremath, 2016; Toure, Maiga & Ouattara 2022). Furthermore, the growing size of the population in a city leads to an increase in the amount of solid waste that is a severe threat to the natural environment and the underground water (MoEF, 2000; Mor *et al.,* 2006 b; Dahiya, 2015). As per an estimate, the current population of Lucknow metropolitan city is about 3,945,000. However, the population of the city has experienced considerable change since the Independence of India in 1947.

The population of Lucknow city was 496177 in 1951 which increased to 8.27 lakhs in 1971 and increased to 28.16 lakhs in 2011 again. The growth rate of the population has experienced a substantial increase from 14.6% in 1971, 22.38% in 1981, 70.79% in 1991, 35% in 2001, and 28.8% in 2011. The highest growth rate was observed between 2001 and 2011, indicating a rapid increase in urbanization (Figure 3).

Similarly, the population growth of the city has also experienced fluctuation since Independence. The population of Lucknow was 496177 in 1951 which increased to 8.27 lakhs in 1971 and reached 28.16 lakhs in 2011 (Census of India, 1951 to 2011). The population growth rate of Lucknow city has experienced a substantial increase trend from 14.6% in 1971, 22.38% in 1981, 70.79% in 1991, 35% in 2001, and 28.8% in 2011. The highest growth rate was observed between 2001 and 2011, indicating a rapid increase in urbanization (Figure 4).

The city's population density has also registered variation in growth since Independence. The population density shows that it decreased from 1971 to 1981 and rose in 1991 and again decreases till 2001; finally, it rose again in 2021. The population density of Lucknow was 10712 people per $Km^2$ in 1971 which rose to 10861 people per $Km^2$ in 2021 (Census of India, 1971 to 2021). The area of the city has experienced a considerable

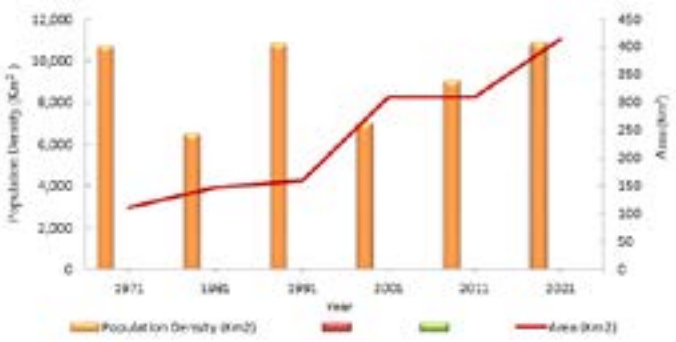


**Figure 4:** Population Density and Area of Lucknow Municipal Corporation, 1971-2021
**[Source:** Master Plan 2021, Census of India]

increase trend from 110 people per $Km^2$ in 1971, 159 $Km^2$ in 1991, 310 people per $Km^2$ in 2001, and 414 people in 2021. Maximum expansion in an urban area has been observed between 2011 and 2021, indicating a rapid increase in urbanization in the city.

The population of the Lucknow urban area including the total population along with the urban fringe of the city has accounted for significant change since 1971. The total population of Lucknow urban area was recorded at 8.10 lakhs in 1971 which rose to 16.70 lakhs in 1991 and 29.00 lakhs in 2011 and finally reached 45.00 lakhs in 2021. The population growth rate of the Lucknow urban area has observed an unexpected trend from 44% in 1981 to 66% in 1991; thereafter its slowdown to reach 40% in 2021. The highest population growth rate has been noted from 1981 to 1991 which indicates towards huge expansion process of urbanization of Lucknow city during the decade of 1991s (Figure 5). Similarly, the estimated population growth is 58.00% in 2021 with a total population of 44.400 lakhs (Patel & Teli, 2019).

The standard density of population for a metropolitan city is 150 people per hectare (125–175 pph). As per the 'Urban Development Plans Formulation and Implementation' (UDPFI) 2015 guideline, the average population density is low in Lucknow city. Lucknow city can provide accommodation to 46.5 lakh people i.e., it can adjust to 18.35 lakhs more population (Figure 6). Currently, migration is continuously increasing the total population in Lucknow city; as it has been observed since the 1980s along with steady natural growth (Patel, Teli, and Purohit, 2019).

**Urbanization**

Urbanization and industrialization lead to constant solid waste generation (Hiremath, 2016). Urbanization directly leads to

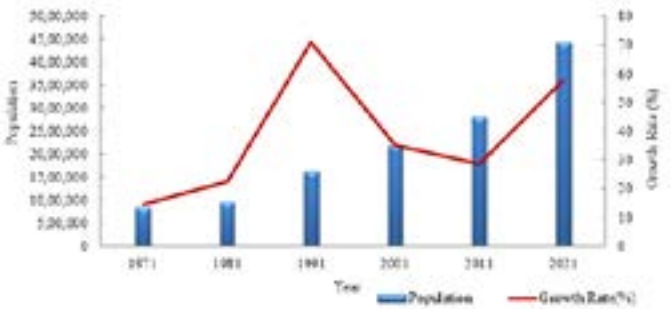


**Figure 3:** Population Growth Rate, Lucknow Municipal Corporation 1971-2021
**[Source:** Master Plan 2021, Census of India]

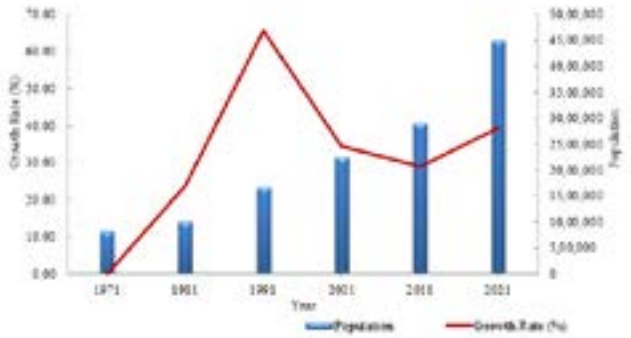


**Figure 5:** Population Growth Rate, Lucknow Urban Area, 1971-2021
**[Source:** Master Plan 2021, Census of India]

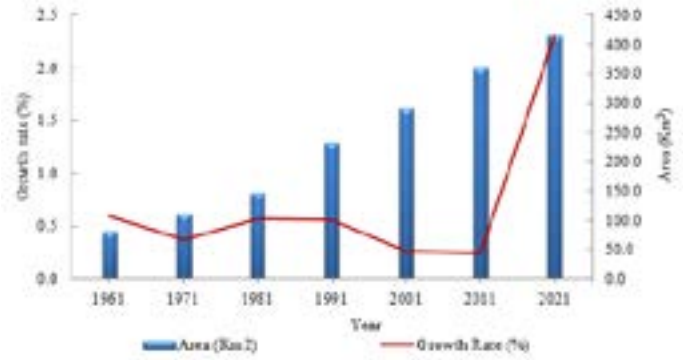



**Figure 6:** Pattern of Urban Growth the City of Lucknow (1961-2021)
**[Source:** Master Plan 2021, Census of India]

produce waste materials and managing irrational waste causes urban environmental degradation and health risks. Solid waste management is a challenging task in our country India is going to be more complex along with rapid urbanization which leads to consumerism and changing lifestyles (Vij, 2012). The mounting urban population is raising the pressure on the surface of the urban areas which results in increasing pressure on the carrying capacity and its utilization.

Before the 1990s, the expansion rate of Lucknow city was manageable; thereafter, the city expanded rapidly, especially during the decade of 1990s. The expansion of CBD of the Lucknow city has increased the outline. The increasing population has accelerated the expansion of the urban area. The number of electoral wards has increased from 87 to 110 in the city. The process of urbanization has registered twice during the 1990s and once again during the 2001s since 1971 (Figure 7). Land utilization for urbanization was cooperatively slow during the 1980s; thereafter 1990s witnessed rapid land consumption, about 4-5$^{th}$ of the total land consumption compared to population growth (Kumari, 2015).

From the above discussion, it is clear that Lucknow city has experienced rapid population growth along with urbanization which has resulted in the city's speedy expansion during the 1980s and 1990s. Next urban growth has been noted between 2011 and 2023 which ultimately extended infrastructure, commercial, residential, industrial areas, etc. in the city. Expansion of Lucknow city is continuing. Resultantly, as a byproduct of city expansion, the rate of solid waste generation is increasing which can be seen mounting the solid wastes at different locations in the city.

**Solid Waste Generation**

Municipal solid waste comprises commercial and residential waste generated within LMC in the form of solid or semi-solid, including treated biomedical wastes except for industrial hazardous wastes. LMC has liable for to collection, transportation, segregation disposal, and processing of solid waste generated within the city. Moreover, LMC manages the analysis and estimate of waste, recourse management, waste reduction, and public awareness in the city. Municipal solid waste generation in Lucknow is about 1365 tons per day with an average per capita solid waste (Kawai, K., Tasaki, T. 2016) of 280 grams per day. Lucknow municipal waste includes organic waste, construction, and demolition waste, street sweeping, recyclable waste, silt waste, mixed waste, etc. Municipal solid waste in Lucknow includes waste generated from residential areas, commercial areas, healthcare systems, hotels & restaurants, street sweepings construction and demolition sector, and drain silt. There are 110 wards where LMC workers and private workers collect solid waste from door-to-door and other sites. Additionally, there are dozens of commercial areas such as Aminabad, Charbag, Alambagh, etc. where waste starts collecting between 8 AM to 2 PM with nominal charges. The biomedical waste generated healthcare sector includes 200 hospitals, 344 medical clinics, 200 nursing homes, 33 pathologies, 93 dental clinics, and more than 2000 doctors' clinics. Furthermore, more than 1000 hotels and restaurants generate 5-6 metric tons of waste per day. C&D waste contains 16% of the total depends on construction or demolition activities in Lucknow city. LMC's rubbish removal (RR) department is responsible for street sweeping activities done mechanically and manually. The street sweeping shares about 3% of the solid waste generated in Lucknow. Besides, drain silt constitutes 6% of the MSW generated including household waste, dust, construction waste, sweeping, etc. Municipal Solid Waste has been classified into major three categories i.e., recyclable, bio-degradable, and inert compositions. The bio-degradable organic solid waste constitutes 47%; C&D waste shares 16%; and recyclable waste shares 17 and 10% of the city's drain & silt and mixed waste (Lucknow Municipal Corporation, 2015).

Projected future population indicates that the total population of Lucknow city is increasing continuously which will reach 38.03 lakhs in 2021 from 28, 17,105 in 2011. Further, the population of the city will rise to 53.59 lakhs in 2031. Overall figure 8 suggests that population of the Lucknow will rise about two times from 2011 which ultimately produces a huge amount of solid waste Figure 8.

The national average of per capita/day solid waste generation in the 28 cities ranges from 0.19 kg to 0.99 kg and the average for all 28 cities is 0.39 kg (at estimated population 2021) (NITI Aayog, 2021). As per MHUF, Govt. of India, per capita/day solid waste generation varies between 0.2 Kg to 0.6 Kg in cities with a population having 1.00 lakh to 50.00 lakhs

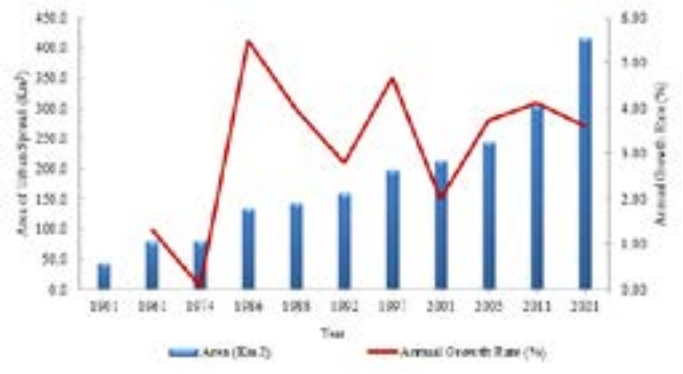



**Figure 7:**Growth of Urban Sprawl, Lucknow City (1901-2021)
**[Source:** Compiled from various historical maps, SOI Toposheet, and Census]

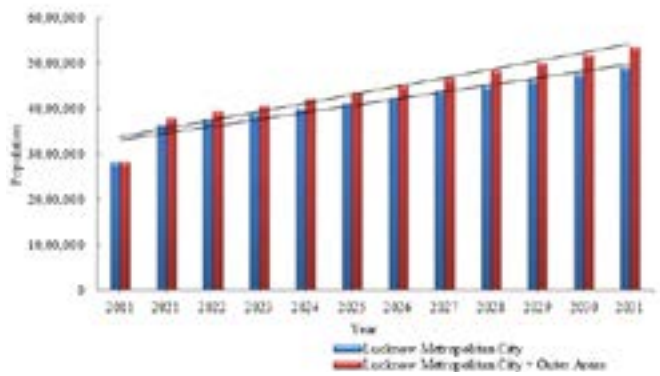



**Figure 8:** Projected Future Population of Lucknow City, 2021-2031
**[Data Source:** United Nations - World Population Prospects]

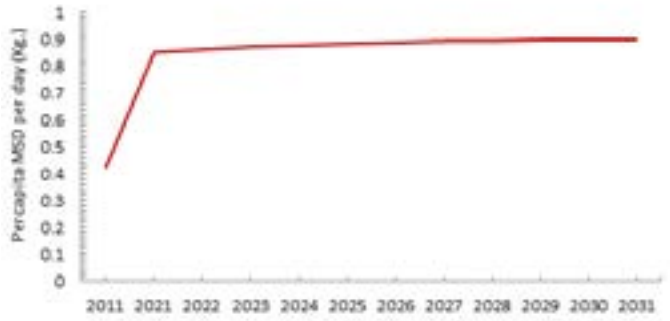



**Figure 9:** Projected Per Capita Municipal Solid Waste Generation Per Day (Kg.) Lucknow (2011-2031)
**[Data Source:** Lucknow Municipal Corporation]

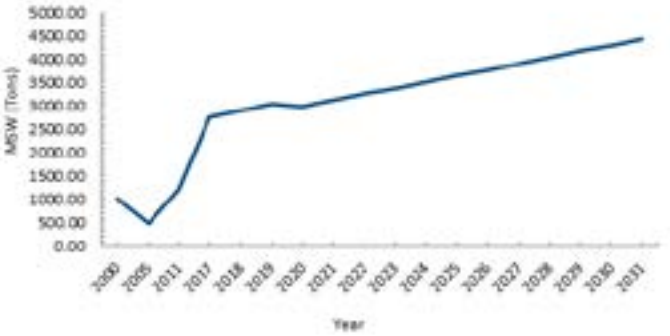



**Figure 10:** Projected Municipal Solid Waste Generation (TPD) Lucknow (2000-2031)
**[Data Source:** World Population Prospects, United Nations]

(Ministry of Housing and Urban Affairs). In Lucknow, per capita per day solid waste generation is relatively low. Figure 8 displays the projected per capita solid waste generation per day that indicates towards sharp increase from 2011 to 2021 and maintains its rate onward in the next decade i.e., the year of the deadline for sustainable development. Figure indicating that per capita municipal solid waste generation will be 0.84 kg every day which will be a challenging task for LMC.

Lucknow city produces about 2,000 tons of solid waste per day (Chauhan, 2022). Figure 10 shows that municipal solid waste generation in Lucknow city was 475 tons per day in 2005 which had increased to 1200 tons per day in 2011. However, the city's estimated municipal solid waste generation was 2989.44 tons per day in 2021 per capita. However, the figure shows that municipal solid waste generation will rise to 4429.05 tons per day in 2031. In brief, the city's projected municipal solid waste generation is continuously mounting its amount during the 2020s decade till 2031 and rose to 4429.05 tons per day (Figure 10).

Figure 8, 9, and 10 displays the projected items i.e., population, municipal waste, and per capita solid waste generation per day that indicates that as per population size of the city is increasing along with area expansion of the city, per capita solid waste generation as well as total municipal solid waste generation is increasing upwards to the year of the SDGs target. There is, therefore, a correlation between the expansion of the city in terms of population and area and the generation of solid waste within the city.

**Sustainable Solid Waste Management System**

The sustainable solid waste management system (SSWMS) is one of the recent and most popular approaches for managing waste generated in an eco-friendly sustainable way. Solid waste loaded with problems of huge quantity is growing across the world. On the other hand, it harms our living environment with the special context in materials dangerous to our ecosystems. An SSWMS is needed for human society to address anxieties from socio-economic and environmental viewpoints. Sustainable development is the social and economic progress of the current generations without compromising the progress of the future generation. The SSWMS is based on the principle of the four R's of reduce, reuse, recycle, and refuse (Korhonen, Nuur, Feldmann & Birkie, 2018) which must be applied to have a sustainable solid waste management system. It needs to increase waste collection efficiency. It was important to discourage and avoid raw and open dumping and promote public participation to keep our clean and green environment. The circular economy (Ellen MacArthur Foundation, 2015a) is the most important element of sustainable waste management that applies an approach to social and economic development aiming to remove growth by utilizing limited resources. It opens the paths of an immediate solution to various problems and addresses the key issues of the consumerist world.

Sustainable solid waste management is a comprehensive eco-friendly solution to control waste generation, dumping, storage, transfer, and disposal in the best method for our health. The four R's rules are 1) Recycling is an eco-friendly waste management solution that protects our environment and makes safe society's well-being; 2) Reduction is reducing or eliminating various materials initially used. Reducing the number of materials is the most important alternative option to the management of waste sensibly because it mitigates the global garbage problems; 3) Reuse is also an eco-friendly solution for solid waste management. It advocates utilizing materials again in their original state and 4) Recover talks about wastes dangerous for our nature that must be recovered through either the reuse or recycling process. These four processes are the best way to handle non-biodegradable or biodegradable waste to be sustainable.

**Policy Implication for Implementation**

We would like to suggest the most suitable and efficient approach to handle the challenges and difficulties of solid waste management which are already proven true as the best solution for solid waste management in different studies. Such an efficient and effective approach must be framed at the policy level and implemented by the state as well as the central government to make a safer place for our city and planet earth. Such approaches are as follows.

*Integrated Solid Waste Management (ISWM)*

Integrated Solid Waste Management is a strategic methodology for sustainable management of all types of solid waste covering all aspects and all sources, including generation, transfer segregation, sorting, recovery, treatment, and disposal in an integrated way, with a focus on optimizing the use of resource efficiency. It is defined "as the best approach of a combination of waste managing techniques to provide a beat solution to the various types of wastes in effective and efficient ways which are environmentally, economically, and socially sustainable". It must be based on the waste hierarchy principle and focus on applying the Four R's for suitably treating solid waste. The ISWM approach requires public participation and collaboration among all the organizations and individuals engaged in managing waste. According to Van de Klundert and Anschütz (2001) concept of integrated solid waste management is based on four principles: i) Equity refers to everyone has a right to access the services of the solid waste management system that protects their environment and health; ii) Effectiveness is the methods for managing wastes which have must meet the

objective of the waste plan and satisfying the public needs; iii) Efficient waste management is equal and effective for creating suitable use of available resources; and iv) Sustainability of the WMS can be attained if the system is eco-friendly suitable to the local situations and can continue enduring use of available natural, economic and human resources. ISWM runs based on the four R's principle of recycling, reuse, reduce, and refuse to have a sustainable solid waste management system.

The ISWM has four functional elements: source reduction, recycling and composting, waste transportation, and landfilling. Source reduction processes include designing for recycling, durable, sustainable products, designing reusable products, refurbishing goods, redesigning goods, reducing food spoilage and waste, and avoiding goods. Source reduction processes include reducing waste handling, transportation, disposal costs, etc. The recycling process includes collecting, categorizing, and recovering reusable and recyclable substances, as well as reprocessing reusable and recyclables to manufacture new goods. Composting is part of recycling organic materials that includes gathering organic waste and transforming it into soil seasonings. Transportation comprises the process of collection of waste from roadside and industries, and also from transfer posts where waste is deposited and reloaded on the trucks for transfer to landfill. Disposal is the process undertaken to manage the waste things that are not recycled i.e., use of combustion and landfills. Landfilling is the most population method of managing waste that must be well-constructed, correctly designed, and scientifically managed.

*Extended Producer Responsibility (EPR)*

Extended Producer Responsibility (EPR) is defined as an environmental protection approach that compelled to the producer of the product accountable for the complete life cycle of the good and particularly for taking back, reprocessing, and final disposal of the manufactured goods (Lindhqvist, 2000; Khetriwal et al., 2009). It is a policy strategy to combine all the environmental costs related to a product throughout its life cycle to the market price of the product waste production per capita (Statistics Canada, 2002). It aims the source of the waste problem to be accountable to the actors who are capable most to eliminate or reduce waste i.e., the producers and consumers (Quinn & Sinclair, 2006). EPR comprises producers assuming physical and/or financial accountability for the waste generated by their products (R. and J. Fenton Sinclair 2001; Institute for Local Self-Reliance, 2000). The EPR initiatives have been noted to produce considerable environmental benefits, such as reducing energy consumption, decreasing the packaging amount, reducing GHG production levels, and dependency on raw materials, litter abatement, and accelerate recycling rate (Quinn & Sinclair, 2006). Now it is clear that integrated taxes will extend social aims of sustainability be attained because policies that permit direct taxes do not extend manufacturers' environmental accountabilities (Quinn & Sinclair, 2006). EPR suggests that producers must take over the responsibility for gathering or taking back used commodities and for sorting and treating wastes for their final recycling. EPR is part of a circular economy that attempts to provide the best solution within a closed system. EPR work on the principles of the Polluter Pays Principle (PPP) i.e., the waste generator and the waste holder must bear all the costs of waste management in a manner that assures protection of the human health and our environment. EPR is a mechanism to support the implementation of the waste hierarchy, and therefore priority: prevention, reuse, and recycling can be applied. Moreover, it motivates to change the manners of all players engaged in the product value chain: manufacturers, consumers, retailers, local authorities, citizens, private and public waste management owners, recyclers, and socio-economic actors.

Integrated solid waste management (ISWM) and extended producer responsibility (EPR) are the best solutions for preventing degradable and non-degradable waste in a circular economy. It must be implemented for the metropolitan city of Lucknow. However, non-degradable waste is the biggest problem because it cannot be decomposed naturally. We need, therefore, a replacement of such products that must be eco-friendly and prospect sustainable development. There are some replacements and options for non-degradable products to reduce, reuse, and recycle waste litter. For example, by reducing plastic use, we can use packaging material made from natural products like jute bags & rope, cloth packaging, and 'plastic' made from wood, etc. One of the major causes of plastic litter, i.e., single-use plastic crockery, water bottles, etc., can be replaced by disposable crockery made from plant residues like leaves plates, earthen cups, etc. The number of tons of plastic litter in Lucknow can be recycled and converted into material for constructing plastic roads; waste plastic eco-bricks for constructing embankments, shelter houses, street furniture**,** walls, etc. can consume thousands of tons of existing non-degradable in our ecosystem. Moreover, printing on recycled waste plastic and using plastic-eating bacteria to reduce the extensive plastic litter in our ecosystem can be an additional good option.

## CONCLUSIONS

Sustainable solid waste management is a holistic approach based on principles of the four R's of the integrated solid waste management system. Available data analysis reveals that the growing population and urbanization expanding urban areas resulted in growing solid waste generation rapidly in Lucknow. Rapid economic growth, increasing population levels, and a community's living standards accelerate the generation rate of solid waste in Lucknow city. It will cross critical until the year of the target, which will be a very difficult task to manage waste. Mismanagement of solid waste causes risks to inhabitants. Knowledge, prevention, mitigation, removal, and behavioral change are the major aspect among the public and shareholders to manage and transform solid wastes into innovative goods useful for people. In a nutshell, we need innovative eco-friendly, sustainable, and 100 percent biodegradable products and the implementation of extended producer responsibility at all levels to prevent and protect waste in our ecosystems.

## CONFLICT OF INTEREST

There is no conflict of interest.

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